LINEAR ALGEBRA · LECTURE 8

Matrix Transformations

Reflection · Projection · Rotation · Scaling & Shearing
Dilation & Contraction · Translation in 2D and 3D



Muhammad Zubair

Lecture notes for BS students | with many pictures, worked examples and a summary sheet

Built on: Anton & Rorres, *Elementary Linear Algebra* (§1.8–1.9) · G. Williams, *Linear Algebra with Applications* (§2.6) · DeFranza & Gagliardi, *Introduction to Linear Algebra with Applications* (§4.6)

Contents

1 The big idea: a matrix moves space **3**

2 Transformations and matrix transformations **3**

2.1 Functions that eat vectors . . . 3

2.2 Matrix transformations . . . 4

3 Linear transformations: the two golden rules **4**

4 The standard matrix: just watch $\mathbf{e}_1$ and $\mathbf{e}_2$ **5**

5 Reflection: the mirror **6**

5.1 Reflections in $\mathbb{R}^2$ . . . 7

5.1.1 Reflection about any line through the origin . . . 7

5.2 Reflections in $\mathbb{R}^3$: mirrors are planes . . . 8

6 Projection: the shadow **9**

6.1 Projections in $\mathbb{R}^2$ . . . 9

6.1.1 Projection onto any line through the origin . . . 9

6.2 Projections in $\mathbb{R}^3$ . . . 10

6.3 From 3D to the screen: parallel projection . . . 10

7 Rotation: the turn **11**

7.1 Rotation in $\mathbb{R}^2$ . . . 11

7.2 Rotation in $\mathbb{R}^3$: about an axis . . . 12

8 Scaling and shearing: stretch and slant **13**

8.1 Scaling . . . 13

8.2 Shearing . . . 14

9 Dilation and contraction: zoom in, zoom out **15**

10 Translation and homogeneous coordinates **16**

10.1 The trick: add a coordinate equal to 1 . . . 16

10.2 Combining moves (read the product from right to left) . . . 17

10.3 Rotating about a point that is not the origin . . . 17

10.4 Translation in 3D: 4×4 matrices . . . 18

11 Composition: doing one move after another **18**

12 Everything on one page **19**

13 Applications **20**

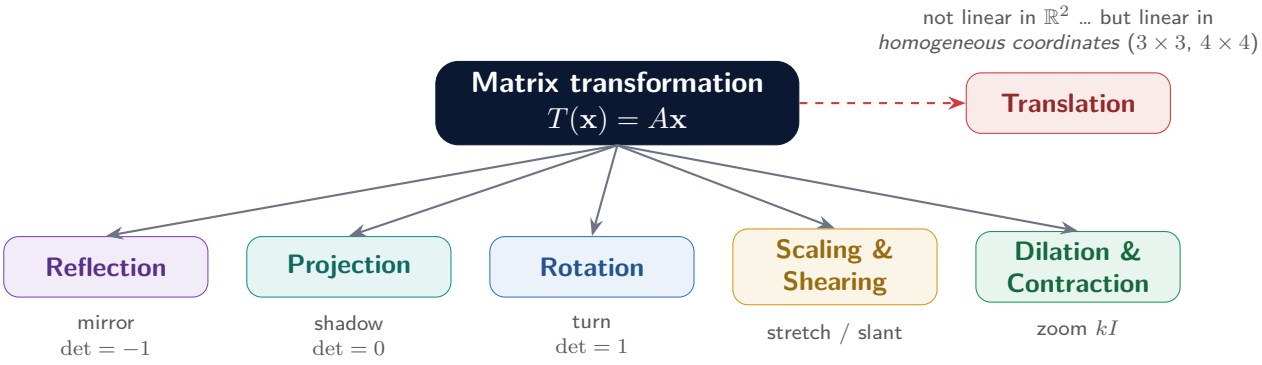
13.1 Computer graphics 20

13.2 Fractal ferns: nature from four matrices 20

14 Check yourself **21**

How to read these notes

- In every picture the **blue** shape is the *original* and the **orange** shape is the *image* after the transformation.
- We use the letter **F** on purpose: it has no symmetry, so you can always see whether a picture was *flipped*, *turned*, *stretched* or *slanted*.
- The **green** arrow is where $\mathbf{e}_1 = (1, 0)$ goes, and the **red** arrow is where $\mathbf{e}_2 = (0, 1)$ goes. These two arrows *are* the columns of the matrix.
- Boxes: Definition | In simple words | Theorem | Example | Common mistake.

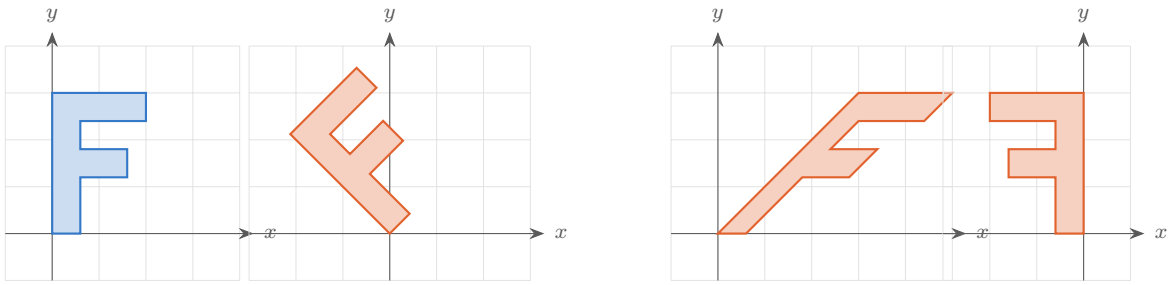


1 The big idea: a matrix moves space

When you rotate a photo on your phone, zoom a map, or play a 3D game, the computer is doing one thing again and again:

take a point $\mathbf{x}$ $\longrightarrow$ multiply by a matrix A $\longrightarrow$ get a new point $A\mathbf{x}$.

A single 2×2 matrix can turn, flip, stretch, squash or slant *every* point of the plane at once. That is what this lecture is about.



Where is this used?

Computer graphics and video games (moving characters and cameras), image editing (rotate, flip, resize), robotics (turning a robot arm), CAD and architecture, physics (changing coordinate systems), machine learning (every layer of a neural network starts with a matrix transformation), and even nature pictures such as fractal ferns (Section 13).

2 Transformations and matrix transformations

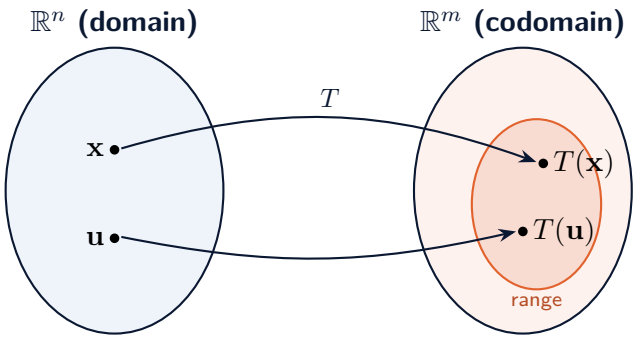
2.1 Functions that eat vectors

Definition Transformation

If T is a function whose domain is $\mathbb{R}^n$ and whose codomain is $\mathbb{R}^m$, we call T a **transformation** from $\mathbb{R}^n$ to $\mathbb{R}^m$ and write $T : \mathbb{R}^n \to \mathbb{R}^m$. If $m = n$, T is also called an **operator** on $\mathbb{R}^n$. For each $\mathbf{x}$, the vector $T(\mathbf{x})$ is the **image** of $\mathbf{x}$. The set of all images is the **range** of T.

In simple words

A transformation is a machine: you put a vector in, a vector comes out. "Domain" is where inputs live, "codomain" is where outputs live.



2.2 Matrix transformations

A linear system

$$\begin{aligned} w_1 &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ &\vdots \qquad\qquad\qquad\qquad \iff \quad \mathbf{w} = A\mathbf{x} \\ w_m &= a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{aligned}$$

can be read in a new way: the $m \times n$ matrix A *sends* each $\mathbf{x} \in \mathbb{R}^n$ to $\mathbf{w} = A\mathbf{x} \in \mathbb{R}^m$.

Definition Matrix transformation

For an $m \times n$ matrix A, the map $T_A : \mathbb{R}^n \to \mathbb{R}^m$, $T_A(\mathbf{x}) = A\mathbf{x}$ is called a **matrix transformation** ("multiplication by A"). When $m = n$ it is a **matrix operator**.

Size rule

A is $m \times n \implies T_A : \mathbb{R}^n \to \mathbb{R}^m$. (**columns** = input size, **rows** = output size.)

Example A transformation from $\mathbb{R}^4$ to $\mathbb{R}^3$ [Anton, §1.8, Example 1]

$w_1 = 2x_1 - 3x_2 + x_3 - 5x_4$, $w_2 = 4x_1 + x_2 - 2x_3 + x_4$, $w_3 = 5x_1 - x_2 + 4x_3$ is multiplication by

$$A = \begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix}, \qquad T_A\left(\begin{bmatrix} 1 \\ -3 \\ 0 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 8 \end{bmatrix}.$$

Two special cases: the **zero transformation** $T_0(\mathbf{x}) = \mathbf{0}$ and the **identity operator** $T_I(\mathbf{x}) = \mathbf{x}$.

3 Linear transformations: the two golden rules

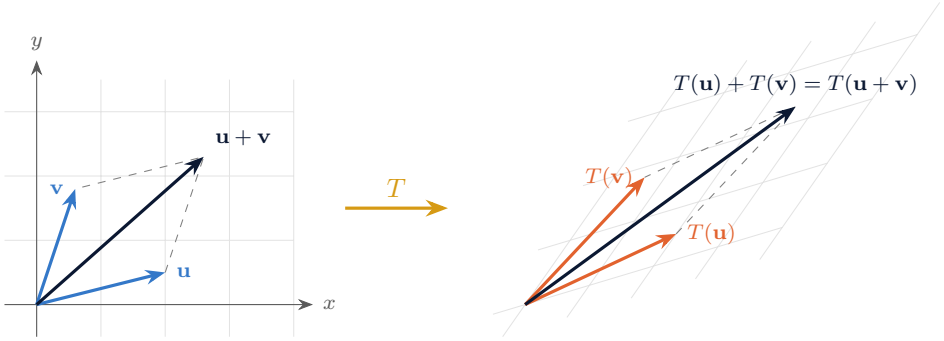
Definition Linear transformation

$T : \mathbb{R}^n \to \mathbb{R}^m$ is **linear** if for all vectors $\mathbf{u}, \mathbf{v}$ and every scalar k:

(i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity) (ii) $T(k\mathbf{u}) = kT(\mathbf{u})$ (homogeneity).

In simple words

A linear map respects "adding" and "scaling": it does not matter whether you add first and then transform, or transform first and then add. Geometrically: straight lines stay straight, parallel lines stay parallel, and the origin stays fixed.



Theorem (Anton 1.8.2–1.8.3)

$T : \mathbb{R}^n \to \mathbb{R}^m$ is a matrix transformation **if and only if** it is linear. So for maps between $\mathbb{R}^n$ and $\mathbb{R}^m$, "linear transformation" and "matrix transformation" mean the same thing.

Quick test (very useful)

Every linear map sends $\mathbf{0}$ to $\mathbf{0}$. So if $T(\mathbf{0}) \neq \mathbf{0}$, then T is **not** linear. (If $T(\mathbf{0}) = \mathbf{0}$ you still have to check the two rules.)

Example Prove linearity [Williams, §2.6, Example 1]

$T(x, y) = (x - y,\ 3x)$. For $\mathbf{u} = (x_1, y_1)$, $\mathbf{v} = (x_2, y_2)$: $T(\mathbf{u}+\mathbf{v}) = (x_1 + x_2 - y_1 - y_2,\ 3x_1 + 3x_2) = (x_1 - y_1, 3x_1) + (x_2 - y_2, 3x_2) = T(\mathbf{u}) + T(\mathbf{v})$, and $T(c\mathbf{u}) = (cx_1 - cy_1, 3cx_1) = cT(\mathbf{u})$. So T is linear; in fact $T(\mathbf{x}) = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix}\mathbf{x}$.

Example Not linear

- $T(x, y, z) = (xy,\ z)$: with $\mathbf{u} = (1, 1, 0)$, $T(2\mathbf{u}) = (4, 0)$ but $2T(\mathbf{u}) = (2, 0)$. [Williams, Example 2]
- $T(x, y) = (x + 2,\ 4y)$: $T(0, 0) = (2, 0) \neq (0, 0)$. A *shift* is never linear in $\mathbb{R}^2$ (this is why translation needs a trick, Section 10).
- $T(x, y) = (x^2,\ y)$: squares break homogeneity.

4 The standard matrix: just watch $\mathbf{e}_1$ and $\mathbf{e}_2$

Every $\mathbf{x} \in \mathbb{R}^2$ is $\mathbf{x} = x\,\mathbf{e}_1 + y\,\mathbf{e}_2$. If T is linear, then

$$T(\mathbf{x}) = x\,T(\mathbf{e}_1) + y\,T(\mathbf{e}_2) = \Big[\, T(\mathbf{e}_1) \;\Big|\; T(\mathbf{e}_2) \,\Big] \begin{bmatrix} x \\ y \end{bmatrix}.$$

Theorem Standard matrix

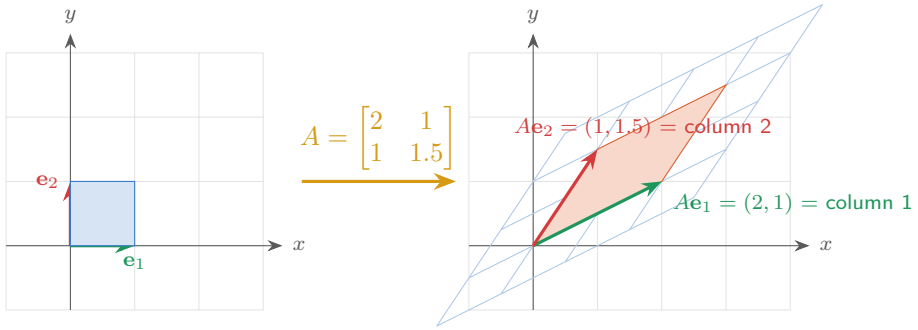
If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear and $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the standard basis vectors, then $T(\mathbf{x}) = A\mathbf{x}$ with

$$A = \Big[\, T(\mathbf{e}_1) \;\Big|\; T(\mathbf{e}_2) \;\Big|\; \cdots \;\Big|\; T(\mathbf{e}_n) \,\Big].$$

A is called the **standard matrix** of T.

In simple words

To find the matrix of *any* linear map: ask "where does $\mathbf{e}_1$ go?" and "where does $\mathbf{e}_2$ go?". Write the two answers as columns. Done.



before after: grid lines stay straight, parallel and evenly spaced

Example Finding a standard matrix [Anton, §1.8, Example 4]

$T(x_1, x_2) = (2x_1 + x_2,\ x_1 - 3x_2,\ -x_1 + x_2)$. Then $T(\mathbf{e}_1) = (2, 1, -1)$ and $T(\mathbf{e}_2) = (1, -3, 1)$, so

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -3 \\ -1 & 1 \end{bmatrix}, \qquad T(1, 4) = A\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ -11 \\ 3 \end{bmatrix}.$$

Example When we only know the images of other vectors [Anton, §1.8, Example 7]

Find A if $T(-1, 1) = (-5, 5)$ and $T(2, -1) = (7, -6)$.
Solution. Write $\mathbf{e}_1 = 1 \cdot (-1, 1) + 1 \cdot (2, -1)$ and $\mathbf{e}_2 = 2 \cdot (-1, 1) + 1 \cdot (2, -1)$. By linearity $T(\mathbf{e}_1) = (-5, 5) + (7, -6) = (2, -1)$ and $T(\mathbf{e}_2) = 2(-5, 5) + (7, -6) = (-3, 4)$, so $A = \begin{bmatrix} 2 & -3 \\ -1 & 4 \end{bmatrix}$.

What the determinant tells you

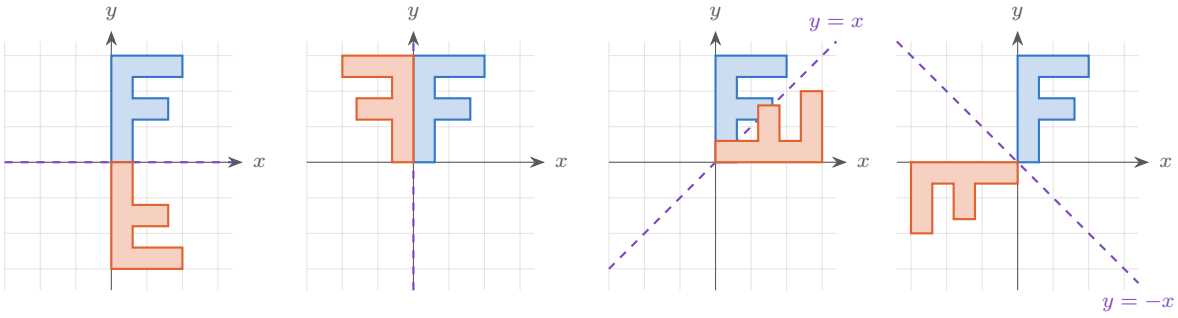
For a 2×2 matrix A, $|\det A|$ = factor by which **areas** change, and the **sign** of $\det A$ tells you whether the picture keeps its orientation $(+)$ or is mirror-flipped $(-)$. If $\det A = 0$ the plane is squashed onto a line or a point (the map is not invertible). In $\mathbb{R}^3$ the same is true for **volumes**.

5 Reflection: the mirror

In simple words

A reflection puts a mirror on a line (in 2D) or on a plane (in 3D). Every point jumps to the other side of the mirror, at the same distance. Points *on* the mirror do not move. Shapes keep their size, but they come out **flipped** (left becomes right).

5.1 Reflections in $\mathbb{R}^2$



about the x-axis $\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$ **about the y-axis** $\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}$ **about the line $y = x$** $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$ **about the line $y = -x$** $\begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix}$

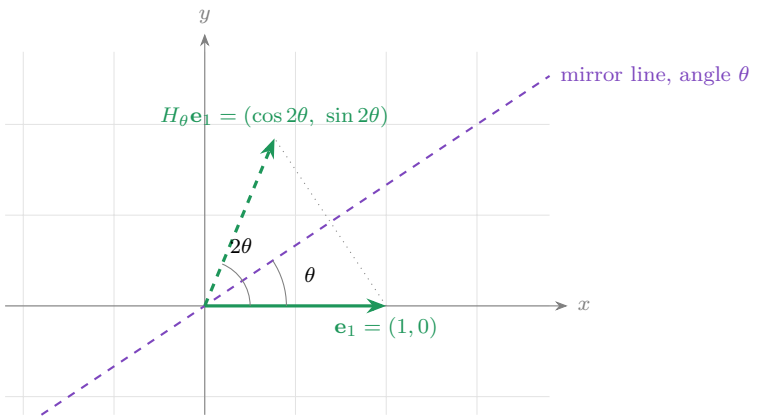
Reflection about	Formula	Images of $\mathbf{e}_1, \mathbf{e}_2$	Standard matrix
x-axis	$(x, y) \mapsto (x, -y)$	$(1, 0),\ (0, -1)$	$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$
y-axis	$(x, y) \mapsto (-x, y)$	$(-1, 0),\ (0, 1)$	$\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}$
line $y = x$	$(x, y) \mapsto (y, x)$	$(0, 1),\ (1, 0)$	$\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$
line $y = -x$	$(x, y) \mapsto (-y, -x)$	$(0, -1),\ (-1, 0)$	$\begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix}$
the origin	$(x, y) \mapsto (-x, -y)$	$(-1, 0),\ (0, -1)$	$\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$

Reflection about any line through the origin

Let the mirror line make an angle θ with the positive x-axis. Reflecting $\mathbf{e}_1$ turns it to the angle 2θ, and reflecting $\mathbf{e}_2$ (at angle $90°$) sends it to the angle $2\theta - 90°$. Hence

$$\boxed{H_\theta = \begin{bmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{bmatrix}}$$

(check: $\theta = 0$ gives the x-axis, $\theta = 45°$ gives $y = x$).



Example **Mirror line $y = 2x$**

Here $\tan\theta = 2$, so $\cos 2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta} = -\dfrac{3}{5}$ and $\sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta} = \dfrac{4}{5}$:

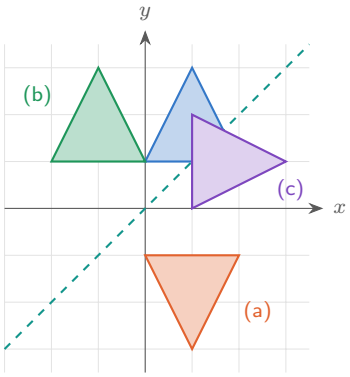
$$H = \frac{1}{5}\begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}. \qquad \text{Check: } H\begin{bmatrix}1\\2\end{bmatrix} = \frac{1}{5}\begin{bmatrix}-3+8\\4+6\end{bmatrix} = \begin{bmatrix}1\\2\end{bmatrix} \text{ (a point on the mirror stays put).}$$

Example **Reflection about $y = -x$** [Williams, §2.6, Example 4]

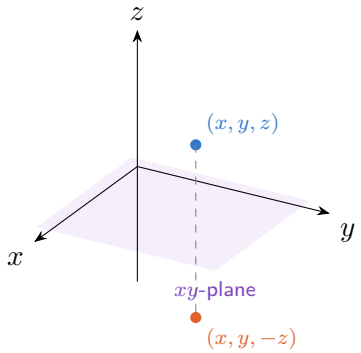
$T(x,y) = (-y,-x)$. $T(\mathbf{e}_1) = (0,-1)$, $T(\mathbf{e}_2) = (-1,0)$, so $A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ and $A\begin{bmatrix}4\\1\end{bmatrix} = \begin{bmatrix}-1\\-4\end{bmatrix}$.

Example **Reflecting a triangle** [DeFranza–Gagliardi, §4.6, Example 3]

Triangle with vertices $\mathbf{v}_1 = (0,1)$, $\mathbf{v}_2 = (2,1)$, $\mathbf{v}_3 = (1,3)$.
(a) x-axis: $(0,-1)$, $(2,-1)$, $(1,-3)$.
(b) y-axis: $(0,1)$, $(-2,1)$, $(-1,3)$.
(c) line $y = x$: $(1,0)$, $(1,2)$, $(3,1)$.
Only the vertices need to be transformed: a linear map sends straight segments to straight segments, so we just join the new vertices.



5.2 Reflections in $\mathbb{R}^3$: mirrors are planes



Mirror	Formula	Matrix
xy-plane	$(x,y,z) \mapsto (x,y,-z)$	$\begin{bmatrix}1&0&0\\0&1&0\\0&0&-1\end{bmatrix}$
xz-plane	$(x,y,z) \mapsto (x,-y,z)$	$\begin{bmatrix}1&0&0\\0&-1&0\\0&0&1\end{bmatrix}$
yz-plane	$(x,y,z) \mapsto (-x,y,z)$	$\begin{bmatrix}-1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

In simple words

To reflect in a coordinate plane, just change the sign of the *one* coordinate that is *not* in the plane's name. (xy-plane: flip z.)

Example [Anton, §1.8, Exercise 43]

For $\mathbf{x} = (1,2,3)$: about xy-plane $\to (1,2,-3)$; about xz-plane $\to (1,-2,3)$; about yz-plane $\to (-1,2,3)$.

Facts about every reflection H

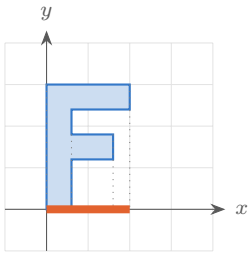
- $H^2 = I$: reflecting twice brings everything back. So $H^{-1} = H$.
- $\det H = -1$: areas (volumes) are kept, orientation is **reversed**.
- Lengths and angles are kept (H is *orthogonal*: $H^T H = I$).
- Two reflections, about the x-axis and then the y-axis, give the reflection about the origin = a rotation by 180° [Anton, §1.9, Example 4].

6 Projection: the shadow

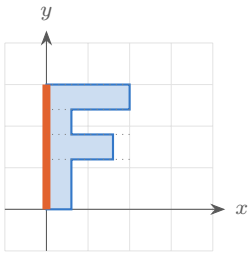
In simple words

An (orthogonal) projection drops every point straight down, at a right angle, onto a line or a plane — like the shadow of an object when the sun is exactly overhead. The shadow is flat: one dimension is lost.

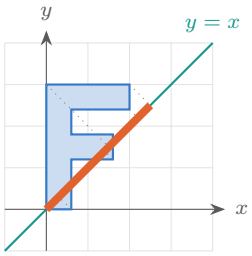
6.1 Projections in $\mathbb{R}^2$



onto the x-axis: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$



onto the y-axis: $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$



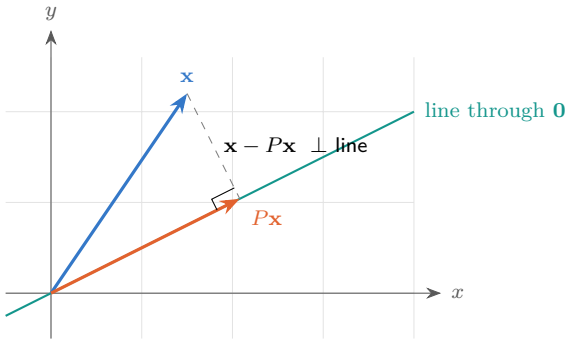
onto $y = x$: $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Projection onto any line through the origin

If the line has unit direction $\mathbf{u} = (\cos\theta, \sin\theta)$, the projection of $\mathbf{x}$ is $(\mathbf{x} \cdot \mathbf{u})\,\mathbf{u}$, so

$$P_\theta = \mathbf{u}\mathbf{u}^T = \begin{bmatrix} \cos^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{bmatrix}$$

For a line along a non-unit vector $\mathbf{a}$: $P = \dfrac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T\mathbf{a}}$.



Example **Projection onto $y = x$** [Williams, §2.6, Exercise 18]

$\theta = 45°$: $P = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ and $P\begin{bmatrix}4\\2\end{bmatrix} = \begin{bmatrix}3\\3\end{bmatrix}$. Indeed $(3,3)$ is the closest point of $y = x$ to $(4,2)$.

Example **Projection onto $y = -x$** [Williams, Chapter 2 Review, Exercise 26]

$\mathbf{a} = (1,-1)$: $P = \frac{1}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$.

6.2 Projections in $\mathbb{R}^3$

Onto	Formula	Matrix	Onto	Formula	Matrix
xy-plane	$(x, y, 0)$	$\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$	x-axis	$(x, 0, 0)$	$\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix}$
xz-plane	$(x, 0, z)$	$\begin{bmatrix}1&0&0\\0&0&0\\0&0&1\end{bmatrix}$	y-axis	$(0, y, 0)$	$\begin{bmatrix}0&0&0\\0&1&0\\0&0&0\end{bmatrix}$
yz-plane	$(0, y, z)$	$\begin{bmatrix}0&0&0\\0&1&0\\0&0&1\end{bmatrix}$	z-axis	$(0, 0, z)$	$\begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}$

Example [Anton, §1.8, Exercise 42]

$\mathbf{x} = (1,2,3)$ projects to $(1,2,0)$ on the xy-plane, $(1,0,3)$ on the xz-plane and $(0,2,3)$ on the yz-plane.

Facts about every projection P

- $P^2 = P$: projecting a shadow again changes nothing.
- $\det P = 0$: **not invertible** — many different points have the same shadow, so you cannot go back.
- Reflection and projection onto the same line are related: $H = 2P - I$ (go to the line, then the same distance again).

6.3 From 3D to the screen: parallel projection

A computer screen is flat, so a 3D object must be *projected* onto the xy-plane. In a **parallel projection** light rays travel in a fixed direction $\mathbf{v}_d = (x_d, y_d, z_d)$ (like sunlight). Following the ray through (x_0, y_0, z_0) until $z = 0$ gives

$$x_p = x_0 - \frac{z_0}{z_d}\,x_d, \qquad y_p = y_0 - \frac{z_0}{z_d}\,y_d \qquad \Longleftrightarrow \qquad \begin{bmatrix}x_p\\y_p\end{bmatrix} = \begin{bmatrix}1 & 0 & -x_d/z_d\\ 0 & 1 & -y_d/z_d\end{bmatrix}\begin{bmatrix}x_0\\y_0\\z_0\end{bmatrix}.$$

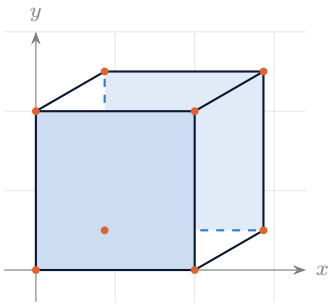
This is a linear map $\mathbb{R}^3 \to \mathbb{R}^2$ (a 2×3 matrix!). [DeFranza–Gagliardi, §4.6, Example 8]

Example Drawing a cube on the screen

With $\mathbf{v}_d = (0.433,\ 0.25,\ -1)$ the matrix is $\begin{bmatrix} 1 & 0 & 0.433 \\ 0 & 1 & 0.25 \end{bmatrix}$. The unit cube's 8 corners go to

$$\begin{aligned} (0,0,0) &\to (0,0) \quad (0,0,1) \to (0.433, 0.25) \\ (1,0,0) &\to (1,0) \quad (1,0,1) \to (1.433, 0.25) \\ (1,1,0) &\to (1,1) \quad (1,1,1) \to (1.433, 1.25) \\ (0,1,0) &\to (0,1) \quad (0,1,1) \to (0.433, 1.25) \end{aligned}$$

Joining the projected corners draws the cube on a flat screen.

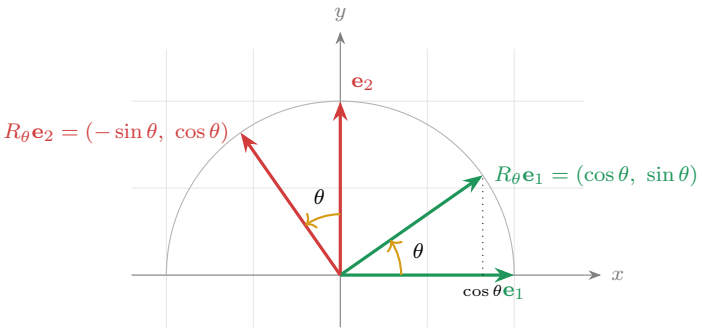


7 Rotation: the turn

In simple words

A rotation turns every point around the origin by the same angle θ. Counterclockwise is positive, clockwise is negative. Nothing is stretched and nothing is flipped — only turned.

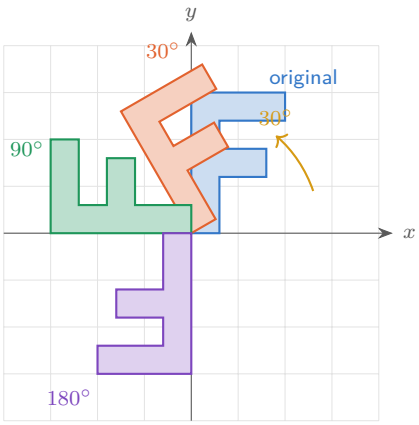
7.1 Rotation in $\mathbb{R}^2$



Reading the two images as columns:

$$\boxed{R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}} \qquad R_\theta \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}.$$

θ	$30°$	$45°$	$90°$	$180°$	$-90°$ (clockwise)
R_θ	$\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$	$\frac{\sqrt{2}}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$



Example A rotation matrix [Anton, §1.8, Example 8]

Rotate $\mathbf{x} = (1, 1)$ through $\pi/6 = 30°$:

$$R_{\pi/6}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{3}}{2} & -\frac{1}{2}\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{3}-1}{2}\\ \frac{1+\sqrt{3}}{2}\end{bmatrix} \approx \begin{bmatrix}0.37\\1.37\end{bmatrix}.$$

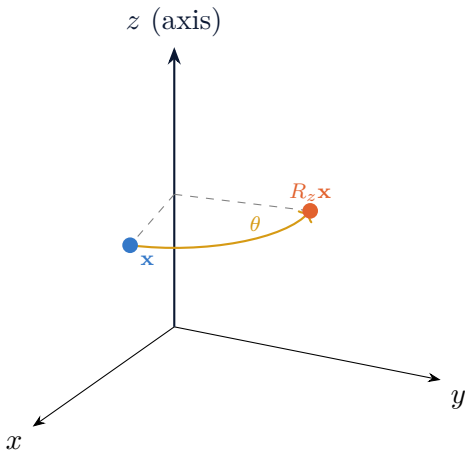
Check: length before $= \sqrt{2} \approx 1.414$, length after $= \sqrt{0.37^2 + 1.37^2} \approx 1.414$. Same, as it must be.

Facts about rotations

- $R_\alpha R_\beta = R_{\alpha+\beta} = R_\beta R_\alpha$: two rotations add their angles, and the order does not matter [Anton, §1.9, Example 3].
- $R_\theta^{-1} = R_{-\theta} = R_\theta^T$: to undo a turn, turn back [Anton, §1.9, Example 6].
- $\det R_\theta = \cos^2\theta + \sin^2\theta = 1$: areas and orientation are kept; lengths and angles are kept.

7.2 Rotation in $\mathbb{R}^3$: about an axis

In 3D we rotate about an *axis*. Use the **right-hand rule**: point your right thumb along the positive axis; your fingers curl in the positive (counterclockwise) direction. The coordinate along the axis does not change; the other two coordinates rotate exactly like in 2D.



about the x-axis	about the y-axis	about the z-axis
$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$	$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$	$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$
x stays; $y \to z$	y stays; $z \to x$ (note the sign!)	z stays; $x \to y$

Example Quarter turns in space

$R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ sends $(1,0,0) \mapsto (0,1,0)$ and $(2,3,5) \mapsto (-3,2,5)$. $R_x(90^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$ sends $(1,1,1) \mapsto (1,-1,1)$.

Common mistake

In $R_y(\theta)$ the minus sign is in the *bottom-left* corner, not the top-right. The cyclic order is $x \to y \to z \to x$: about the y-axis, z turns towards x.

8 Scaling and shearing: stretch and slant

8.1 Scaling

Definition Scaling

$S = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ $(a, b > 0)$ multiplies x by a and y by b: $(x, y) \mapsto (ax, by)$. *Horizontal* scaling: $\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$; *vertical* scaling: $\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$. If $k > 1$ it **stretches**, if $0 < k < 1$ it **compresses**. When $a \neq b$ shapes get distorted (a circle becomes an ellipse).

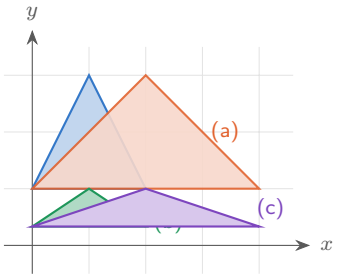
Example Stretch and squash a triangle [DeFranza–Gagliardi, §4.6, Example 1]

Triangle $\mathbf{v}_1 = (0,1)$, $\mathbf{v}_2 = (2,1)$, $\mathbf{v}_3 = (1,3)$.

(a) Horizontal stretch by 2, $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$: $(0,1)$, $(4,1)$, $(2,3)$.

(b) Vertical contraction by 3, $\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$: $(0, \frac{1}{3})$, $(2, \frac{1}{3})$, $(1,1)$.

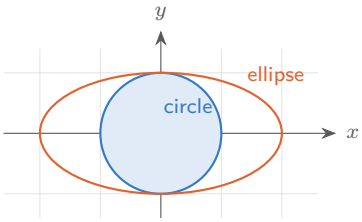
(c) Both: $\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$: $(0, \frac{1}{3})$, $(4, \frac{1}{3})$, $(2,1)$.



Example A circle becomes an ellipse

Scale by 2 horizontally and 1 vertically: $X = 2x$, $Y = y$. Substituting $x = X/2$, $y = Y$ into

$x^2 + y^2 = 1$ gives $\dfrac{X^2}{4} + Y^2 = 1$, an ellipse. The area changes from π to 2π: $\det S = 2$.



8.2 Shearing

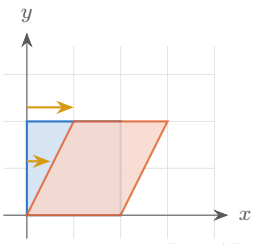
Definition Shear

A **shear in the x-direction** with factor k: $(x, y) \mapsto (x + ky,\ y)$, matrix $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$.

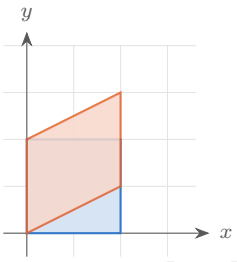
A **shear in the y-direction** with factor k: $(x, y) \mapsto (x,\ y + kx)$, matrix $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$.

In simple words

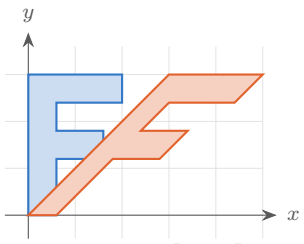
Think of a deck of cards lying on a table. Push the top of the deck sideways: every card slides, and higher cards slide more. Points on the x-axis ($y = 0$) do not move; the higher the point, the further it moves. The height and the area do not change.



x-shear, $k = \frac{1}{2}$: $\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$



y-shear, $k = \frac{1}{2}$: $\begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$



the F under $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Example Horizontal shear of a triangle [DeFranza–Gagliardi, §4.6, Figure 7]

Triangle $(0,0), (2,0), (1,1)$ and $k = 2$: $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ gives $(0,0),\ (2,0),\ (1 + 2 \cdot 1,\ 1) = (3,1)$.

Example Vertical shear [DeFranza–Gagliardi, §4.6, Example 2]

Triangle $(1,1),\ (2,1),\ (\frac{3}{2}, 3)$ and $k = 2$: $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ gives $(1,3),\ (2,5),\ (\frac{3}{2}, 6)$. Each y increases by $2x$.

Example What happens to a line?

Apply the x-shear with $k = 1$ to the line $y = x$. A point (t, t) goes to $(t + t,\ t) = (2t, t)$, which lies on $Y = \frac{1}{2}X$. Lines go to lines — only the slope changed.

Facts

$\det \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = ab$ (areas multiply by ab). $\det \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} = 1$: a shear **keeps area** (same base, same height). Both are invertible: $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}^{-1} = \begin{bmatrix} 1/a & 0 \\ 0 & 1/b \end{bmatrix}$, $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -k \\ 0 & 1 \end{bmatrix}$.

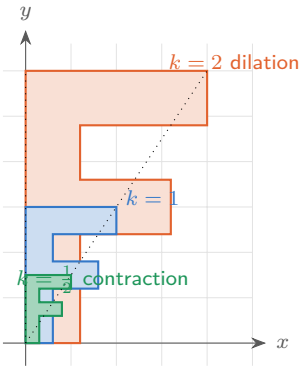
9 Dilation and contraction: zoom in, zoom out

Definition Dilation / contraction

For a scalar $k \geq 0$ the operator $T(\mathbf{x}) = k\mathbf{x}$ has matrix kI:

$$\mathbb{R}^2: \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \qquad \mathbb{R}^3: \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}.$$

If $k > 1$ it is a **dilation** (everything is pushed away from the origin), if $0 \leq k < 1$ it is a **contraction** (everything is pulled towards the origin). It is a *uniform* scaling: the shape stays the same, only the size changes.



Facts

Lengths are multiplied by $|k|$, areas by k^2 ($\det = k^2$ in $\mathbb{R}^2$), volumes by k^3 ($\det = k^3$ in $\mathbb{R}^3$). Angles and shapes are kept. kI commutes with *every* matrix, so a dilation can be done before or after any other transformation.

Example Expansion of space [Williams, §2.6, Exercise 29]

Every point of $\mathbb{R}^3$ moves three times as far from the origin: $T(\mathbf{x}) = 3\mathbf{x}$, matrix $3I_3$; e.g. $(1, -2, 4) \mapsto (3, -6, 12)$; volumes become 27 times bigger.

Example Rotate and zoom at the same time

$2R_{90^\circ} = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$ turns every point by 90° *and* doubles its distance from the origin: $(1, 1) \mapsto (-2, 2)$. Repeating such a map draws a beautiful spiral.

10 Translation and homogeneous coordinates

In simple words

A translation slides every point by the same amount: "3 units right and 1 unit up". It is the most natural move of all, yet it is **not** a matrix transformation of $\mathbb{R}^2$ — because it moves the origin.

Definition Translation

Translation by a fixed vector $\mathbf{b} = (h, k)$ is $S(\mathbf{v}) = \mathbf{v} + \mathbf{b}$, i.e. $(x, y) \mapsto (x + h,\ y + k)$. Since $S(\mathbf{0}) = \mathbf{b} \neq \mathbf{0}$, it is **not linear** (unless $\mathbf{b} = \mathbf{0}$), so no 2×2 matrix can do it.

10.1 The trick: add a coordinate equal to 1

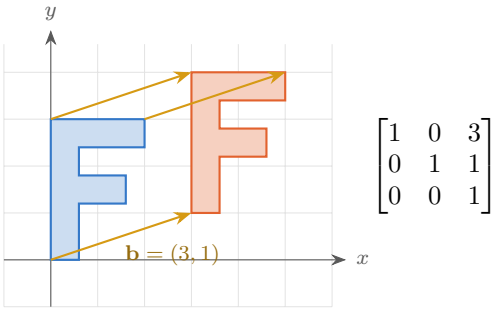
Write the point (x, y) as the column $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$ (its **homogeneous coordinates**). Then

$$\boxed{\begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + h \\ y + k \\ 1 \end{bmatrix}}$$

— a translation done by *matrix multiplication*! Every 2×2 matrix A also fits in this system as $\begin{bmatrix} A & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$, for example

$$\underbrace{\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{rotation}} \quad \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{reflection in } x\text{-axis}} \quad \underbrace{\begin{bmatrix} r & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{dilation}} \quad \underbrace{\begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{shear}}.$$

Now *every* move, including translation, is a 3×3 matrix, and a whole sequence of moves is **one** matrix (their product). This is exactly how graphics software works.



$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Example Translate a point [DeFranza–Gagliardi, §4.6]

Translate $\mathbf{v} = (3, 2)$ by $\mathbf{b} = (1, -2)$: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$, so $S(\mathbf{v}) = (4, 0)$. To undo it, translate by $-\mathbf{b}$: $\begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix}$.

10.2 Combining moves (read the product from right to left)

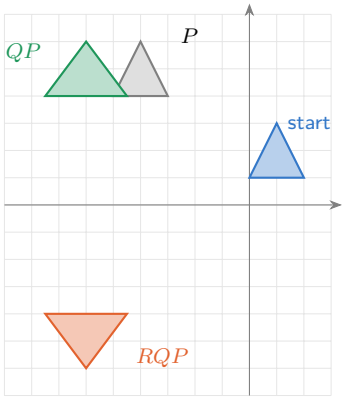
Example **Translate, scale, reflect** [DeFranza–Gagliardi, §4.6, Example 5]

Triangle $(0,1),(2,1),(1,3)$: first translate by $(-5,3)$ (P), then scale horizontally by 1.5 (Q), then reflect in the x-axis (R):

$$RQP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1.5 & 0 & -7.5 \\ 0 & -1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Images: $(0,1) \mapsto (-7.5,-4)$,
$(2,1) \mapsto (-4.5,-4)$, $(1,3) \mapsto (-6,-6)$.

Intermediate steps:
after P: $(-5,4),(-3,4),(-4,6)$
after Q: $(-7.5,4),(-4.5,4),(-6,6)$
after R: $(-7.5,-4),(-4.5,-4),(-6,-6)$.



10.3 Rotating about a point that is not the origin

A rotation matrix always turns around the origin. To turn around a point $P(h,k)$:

(1) slide P to the origin $\rightarrow$ **(2)** rotate $\rightarrow$ **(3)** slide back.

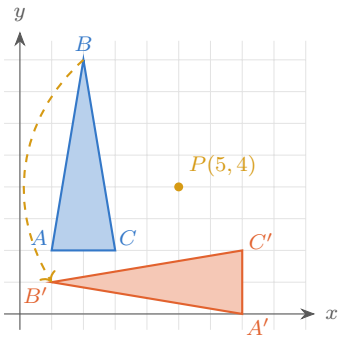
$$R_P = \begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & -h\cos\theta + k\sin\theta + h \\ \sin\theta & \cos\theta & -h\sin\theta - k\cos\theta + k \\ 0 & 0 & 1 \end{bmatrix}.$$

Example **Quarter turn about** $(5,4)$ [Williams, §2.6, Example 5]

$h = 5,\ k = 4,\ \theta = \pi/2$: $\quad M = \begin{bmatrix} 0 & -1 & 9 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}$.

$A(1,2) \mapsto A'(7,0)$
$B(2,8) \mapsto B'(1,1)$
$C(3,2) \mapsto C'(7,2)$.

Check: $P = (5,4)$ itself does not move:
$M(5,4,1)^T = (5,4,1)^T$.



Example **Scaling while one vertex stays fixed** [DeFranza–Gagliardi, §4.6, Example 6]

To stretch by 3 horizontally and 2 vertically while keeping the vertex $(1,1)$ fixed: translate $(1,1)$ to the origin, scale, translate back:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & -2 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

10.4 Translation in 3D: 4×4 matrices

In space, (x, y, z) becomes $(x, y, z, 1)$ and

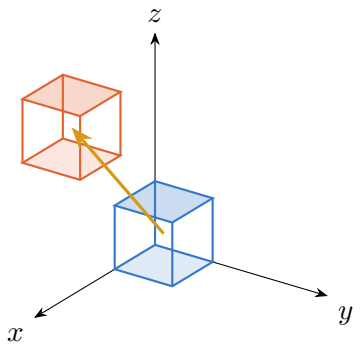
$$\begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+a \\ y+b \\ z+c \\ 1 \end{bmatrix}, \qquad \text{any } 3 \times 3 \text{ matrix } A \text{ becomes } \begin{bmatrix} A & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}.$$

Example **Move a box in space**

Translate by $(2, -1, 4)$: the corner $(1, 2, 3)$ goes to $(3, 1, 7)$.

Rotate 90° about the z-axis *and then* translate by $(2, -1, 4)$: one matrix

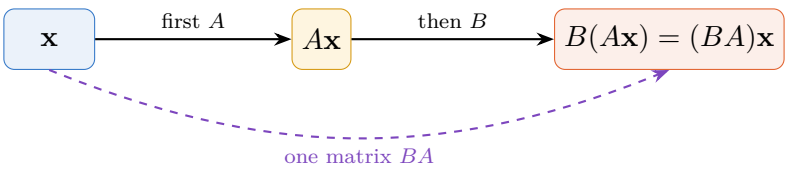
$$\begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (1, 2, 3) \mapsto (0, 0, 7).$$



11 Composition: doing one move after another

Theorem **(Anton 1.9.1)**

If T_A is applied first and T_B second, then $T_B \circ T_A = T_{BA}$: the combined matrix is BA (**right to left**: the first move is written last).



Example **Three moves in a row** [Anton, §1.9, Example 5]

Rotate by $\pi/6$ (A), then reflect in $y = x$ (B), then project onto the y-axis (C):

$$CBA = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}, \qquad CBA \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{\sqrt{3}}{2}x - \frac{1}{2}y \end{bmatrix}.$$

Example **Order matters** [Anton, §1.9, Example 2]

A = reflection in $y = x$, B = projection onto the y-axis: $AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ but $BA = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.

Different matrices, different results.

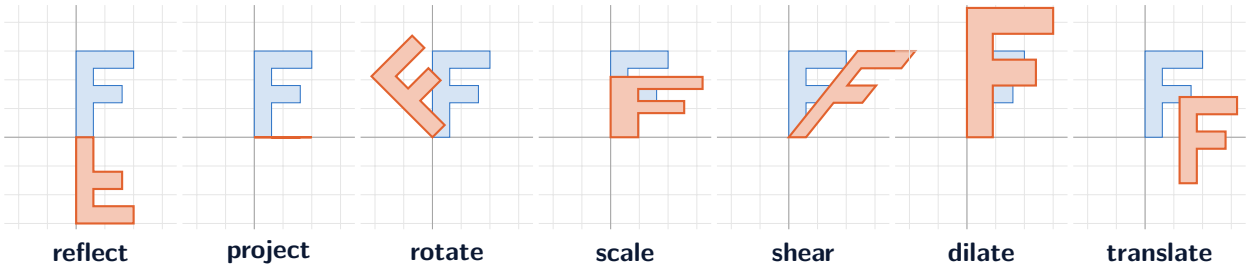
Pair	Commute?	Reason
rotation & rotation	yes	angles add: $R_\alpha R_\beta = R_{\alpha+\beta}$
dilation kI & anything	yes	$kI\,M = M\,kI$
reflection in x-axis & in y-axis	yes	both give $-I$ [Anton §1.9 Ex. 4]
reflection & projection	usually no	e.g. $y = x$ and y-axis above
rotation & shear / non-uniform scaling	usually no	check AB vs BA
translation & rotation	no	rotating moves the translated object

Undoing a sequence

$(BA)^{-1} = A^{-1}B^{-1}$: undo the moves in the **reverse order** (like taking off socks and shoes). Example [DeFranza–Gagliardi, §4.6, Example 4]: reflect in $y = x$ then stretch horizontally by 2: $S = \begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}0 & 2\\ 1 & 0\end{bmatrix}$, $S^{-1} = \begin{bmatrix}0 & 1\\ \frac{1}{2} & 0\end{bmatrix}$.

12 Everything on one page

Transformation	Matrix ($\mathbb{R}^2$)	det	Invertible	Lengths kept	Picture effect
Reflection in x-axis	$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$	-1	yes	yes	mirror flip
Reflection in line at θ	$\begin{bmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{bmatrix}$	-1	yes	yes	mirror flip
Projection on x-axis	$\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$	0	**no**	no	flattened shadow
Projection on line at θ	$\begin{bmatrix}\cos^2\theta & \sin\theta\cos\theta\\ \sin\theta\cos\theta & \sin^2\theta\end{bmatrix}$	0	**no**	no	flattened shadow
Rotation by θ	$\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$	1	yes	yes	turned
Scaling	$\begin{bmatrix}a & 0\\ 0 & b\end{bmatrix}$	ab	if $a, b \neq 0$	if $\lvert a\rvert = \lvert b\rvert = 1$	stretched / squashed
Shear (x-dir.)	$\begin{bmatrix}1 & k\\ 0 & 1\end{bmatrix}$	1	yes	no	slanted, same area
Dilation / contraction	$\begin{bmatrix}k & 0\\ 0 & k\end{bmatrix}$	k^2	if $k \neq 0$	if $k = 1$	zoom
Translation (homog.)	$\begin{bmatrix}1 & 0 & h\\ 0 & 1 & k\\ 0 & 0 & 1\end{bmatrix}$	1	yes	yes	slid



Three questions that identify any 2×2 map

1. Where do $\mathbf{e}_1$ and $\mathbf{e}_2$ go? (the columns) **2.** What is $\det A$? (area factor, flip or not) **3.** Does $A^T A = I$? (if yes: only rotations and reflections — lengths are kept)

13 Applications

13.1 Computer graphics

A game character is stored as a list of vertices (points). To move it on the screen, the program multiplies *all* vertices by one 3×3 (in 2D) or 4×4 (in 3D) homogeneous matrix: scale $\rightarrow$ rotate $\rightarrow$ translate, then project onto the screen (Section 6.3). Because only vertices are moved, and lines go to lines, millions of triangles can be drawn every second.

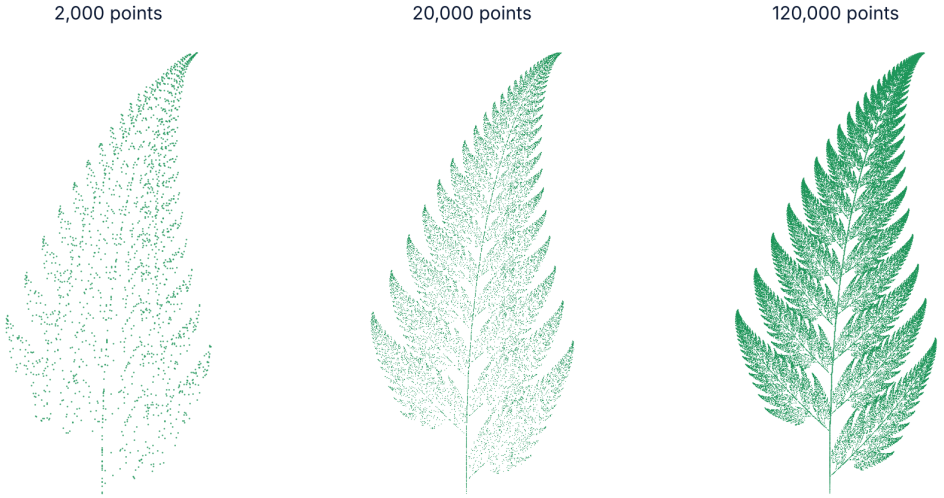
13.2 Fractal ferns: nature from four matrices

Benoit Mandelbrot: *"Clouds are not spheres, mountains are not cones, coastlines are not circles."* Yet a realistic fern can be drawn with just four *affine* maps $T_i(\mathbf{x}) = A_i\mathbf{x} + \mathbf{b}_i$ [Williams, §2.6]:

$$T_1 : \begin{bmatrix} 0.86 & 0.03 \\ -0.03 & 0.86 \end{bmatrix}, \begin{bmatrix} 0 \\ 1.5 \end{bmatrix}, \; p_1 = 0.83 \qquad T_2 : \begin{bmatrix} 0.2 & -0.25 \\ 0.21 & 0.23 \end{bmatrix}, \begin{bmatrix} 0 \\ 1.5 \end{bmatrix}, \; p_2 = 0.08$$

$$T_3 : \begin{bmatrix} -0.15 & 0.27 \\ 0.25 & 0.26 \end{bmatrix}, \begin{bmatrix} 0 \\ 0.45 \end{bmatrix}, \; p_3 = 0.08 \qquad T_4 : \begin{bmatrix} 0 & 0 \\ 0 & 0.17 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \; p_4 = 0.01$$

Start at $(0, 0)$; again and again pick one T_i at random (with probability p_i), apply it, and plot the new point.



14 Check yourself

Check yourself

1. Is $T(x, y) = (x + 1,\ y)$ linear? No: $T(0,0) = (1,0) \neq (0,0)$.
2. Write the matrix that reflects in $y = -x$. $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
3. Which matrix squashes the plane onto the y-axis? $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
4. Rotate $(0, 2)$ by $90°$. $(-2, 0)$
5. What is the area of the unit square after $\begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$? 1 (a shear keeps area)
6. A 2×2 map has $\det A = -6$. What happens to areas and orientation? areas $\times 6$, flipped
7. Write the 3×3 matrix that translates by $(-2, 5)$. $\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$
8. First rotate by $90°$, then reflect in the x-axis. One matrix? $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
9. Which transformations in this lecture are *not* invertible? projections (and kI with $k = 0$)
10. Which 3×3 matrix rotates space by $90°$ about the x-axis? $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

Worked solutions to the textbook exercises are in the separate file *Matrix Transformations — Solved Exercises*.